

MULTIPLE LINEAR REGRESSION

DATA : mtcars

Solution :-

```
# Step 1 : Loading data
data <- mtcars

# Step 2 : Defining dependent and independent variables
y <- data$mpg
x1 <- data$cyl
x2 <- data$disp
x3 <- data$hp
x4 <- data$wt
x5 <- data$gear
x6 <- data$carb

# Step 3 : Creating Design Matrix (add intercept column)
X <- cbind(1, x1, x2, x3, x4, x5, x6)
X <- as.matrix(X)
Y <- as.matrix(y)

# Step 4 : Estimating coefficients manually
# beta_hat = (X'X)^(-1) X'Y
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat

##           [,1]
## 34.476617261
## x1 -0.930807082
## x2  0.008806277
## x3 -0.022031781
## x4 -3.238293032
## x5  1.099495944
## x6 -0.377824105

# Step 5 : Fiting model using lm() for verification
model <- lm(mpg ~ cyl + disp + hp + wt + gear + carb, data = data)
summary(model)

##
## Call:
## lm(formula = mpg ~ cyl + disp + hp + wt + gear + carb, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.3628 -1.6085 -0.4515  1.5206  5.8747
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 34.476617   8.240582   4.184 0.000309 ***
## cyl        -0.930807   0.810740  -1.148 0.261797
## disp         0.008806   0.015804   0.557 0.582332
## hp          -0.022032   0.020286  -1.086 0.287810
## wt          -3.238293   1.406887  -2.302 0.029960 *
## gear         1.099496   1.328284   0.828 0.415641
## carb        -0.377824   0.731295  -0.517 0.609941
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.576 on 25 degrees of freedom
## Multiple R-squared:  0.8527, Adjusted R-squared:  0.8173
## F-statistic: 24.12 on 6 and 25 DF,  p-value: 2.93e-09

# Step 6 : Predicted values
y_hat <- X %*% beta_hat

# Step 7 : Residuals
residuals <- Y - y_hat

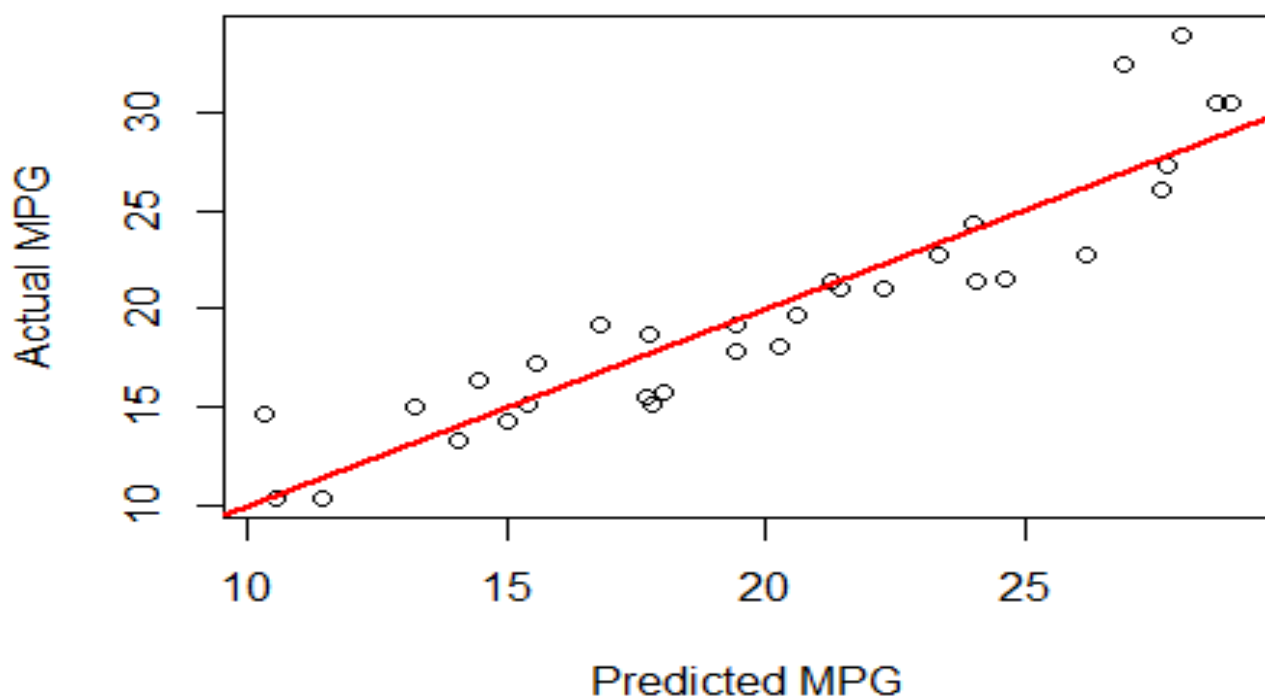
# Step 8 : Calculating R-square manually
SST <- sum((Y - mean(Y))^2)
SSE <- sum(residuals^2)
SSR <- SST - SSE
R2  <- SSR / SST
R2

## [1] 0.8526734

# Step 9 : Ploting (Actual vs Predicted)
plot(y_hat, Y,
     xlab = "Predicted MPG",
     ylab = "Actual MPG",
     main = "Multiple Linear Regression Fit")

abline(0, 1, col="red", lwd=2)
```

Multiple Linear Regression Fit



Data : Trees

Solutions :-

Step 1 : Loading dataset

```
data <- trees
```

Step 2 : Defining variables

```
y <- data$Volume
```

```
x1 <- data$Girth
```

```
x2 <- data$Height
```

Step 3 : Creating Design Matrix (add intercept)

```
X <- cbind(1, x1, x2)
```

```
X <- as.matrix(X)
```

```
Y <- as.matrix(y)
```

Step 4 : Estimating coefficients manually

$\beta_{\text{hat}} = (X'X)^{-1} X'Y$

```
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
```

```
beta_hat
```

```
##           [,1]
```

```
## -57.9876589
```

```
## x1  4.7081605
```

```
## x2  0.3392512
```

Step 5 : Fit model using lm() (verification)

```
model <- lm(Volume ~ Girth + Height, data = data)
```

```
summary(model)
```

```
##
```

```
## Call:
```

```
## lm(formula = Volume ~ Girth + Height, data = data)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -6.4065 -2.6493 -0.2876  2.2003  8.4847
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -57.9877      8.6382  -6.713 2.75e-07 ***
```

```
## Girth        4.7082      0.2643  17.816 < 2e-16 ***
```

```
## Height       0.3393      0.1302   2.607  0.0145 *
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 3.882 on 28 degrees of freedom
```

```
## Multiple R-squared:  0.948, Adjusted R-squared:  0.9442
```

```
## F-statistic: 255 on 2 and 28 DF, p-value: < 2.2e-16
```

Step 6 : Predicted values

```
y_hat <- X %*% beta_hat
```

Step 7 : Residuals

```
residuals <- Y - y_hat
```

Step 8 : Calculate R-square manually

```
SST <- sum((Y - mean(Y))^2)
```

```
SSE <- sum(residuals^2)
```

```

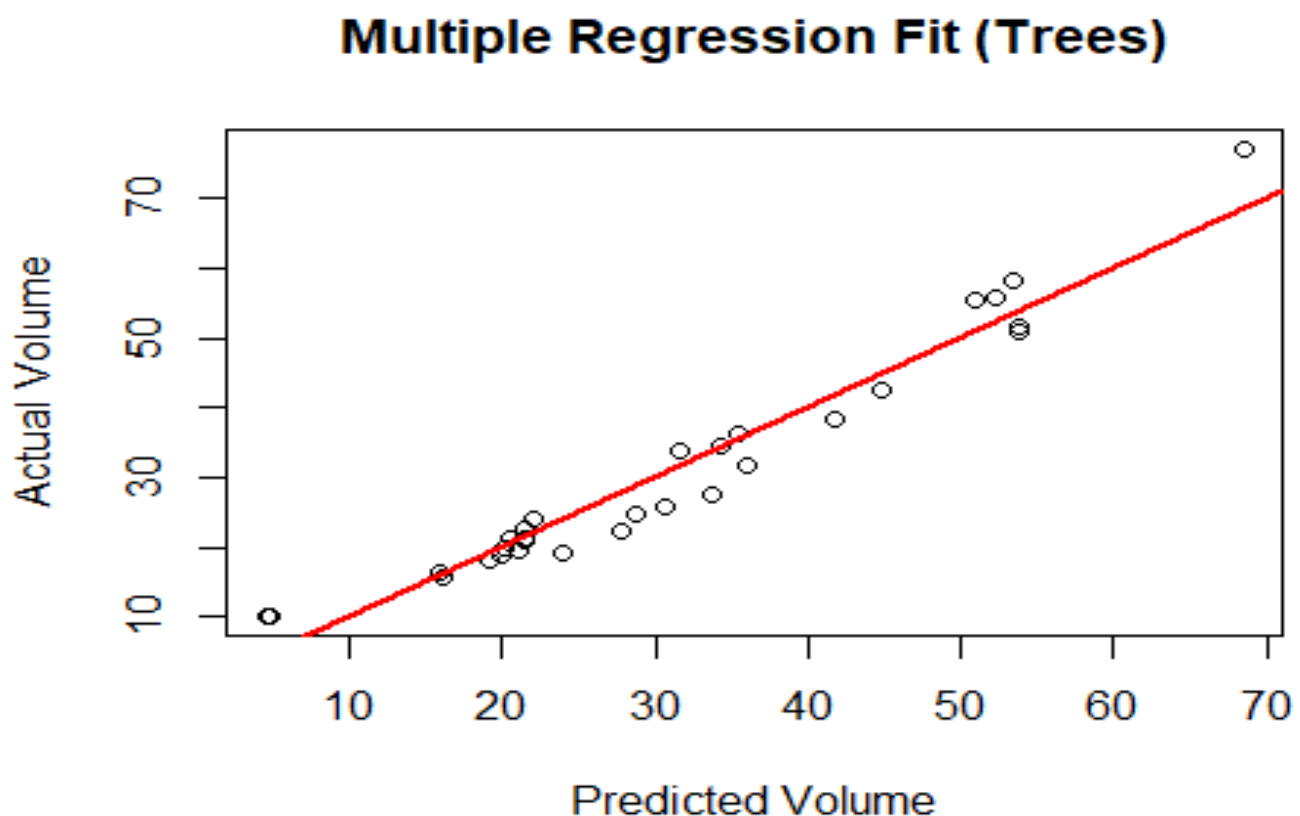
SSR <- SST - SSE
R2 <- SSR / SST
R2

## [1] 0.94795

# Step 9 : Plot (Actual vs Predicted)
plot(y_hat, Y,
     xlab = "Predicted Volume",
     ylab = "Actual Volume",
     main = "Multiple Regression Fit (Trees)")

abline(0, 1, col="red", lwd=2)

```



Interpretation :

Trees dataset:

Tree volume increases as both **girth and height increase**, meaning thicker and taller trees contain more wood. The model shows these two variables strongly explain tree volume.

Mtcars dataset:

Car mileage decreases as **weight, engine size, cylinders, and horsepower increase**, while more gears slightly improve mileage. Heavier and more powerful cars give lower fuel efficiency.